

The pair singular series of a polynomial V : the dilated pieces below $H^{1/2}$

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Abstract

[Draft. The proofs of Theorems 5 and 7 are in [research/explore/PAPER-V-thmM.md](#) and [PAPER-V-smallend-draft.md](#) and are under adversarial reading; nothing here is to be circulated before those readings are recorded.] For a monic irreducible quadratic f without fixed prime divisor, part III decomposed the Cesàro off-diagonal $\text{Off}_f^*(H)$ into pieces indexed by the dilation u and proved that the pieces with $u > H^{1/2+\varepsilon}$ contribute a main term $c_{\text{off}}(f)H$. This paper treats dilations below that threshold. We prove that the pieces with $H^{0.47} < u \leq H^{0.53}$ contribute $O(H^{1-\delta})$, so that $\sum_{u > H^{0.47}} w(u) \mathcal{P}_u(H/u) = c_{\text{off}}(f)H + O(H^{1-\delta})$; the input is a bilinear estimate for Kloosterman fractions of Bourgain and Garaev together with a bound for the additive energy of reciprocals, valid for every modulus, which we deduce from an argument of Cilleruelo and Garaev. For negative discriminants we prove a second result at the other end: the pieces with $u \leq H^{1/3-c}$ contribute $O(H^{1-\delta(c)})$, by running the automorphic kernel method of Grimmelt and Merikoski with the leading coefficient u^2 tracked. What remains is a neighbourhood of $u = H^{1/3}$.

1 Introduction

[Working draft: the proofs are in [research/explore/PAPER-V-thmM.md](#) and [PAPER-V-smallend-draft.md](#) and are under adversarial reading.]

The problem. Let $f = t^2 + bt + c \in \mathbb{Z}[t]$ be monic, irreducible, without fixed prime divisor, of discriminant D . Part III [6] wrote the Cesàro off-diagonal of the pair singular series as a sum of *pieces*,

$$\text{Off}_f^*(H) = P(1) \sum_u \lambda(u) \omega(u) \mathcal{P}_u(H/u),$$

one for each squarefree *dilation* u , the piece $\mathcal{P}_u(Y)$ being the same problem at length $Y = H/u$ with the roots of $x^2 \equiv D$ replaced by those of $u^2 x^2 \equiv D$. Each piece is $O(Y)$, and the weights satisfy $\sum_{u \leq H} |w(u)|/u \asymp \log H$: the trivial bound for the whole sum is $H \log H$, which is exactly the size of the statement to be proved. Every piece must therefore be small, and the difficulty is uniformity in u .

Exponent coordinates. Write $u = H^\alpha$. Each method available for the pieces gives a saving $H^{-s(\alpha)}$ over the trivial bound on an interval of α , and degenerates at the ends of that interval. In these coordinates part III proves $s > 0$ for $\alpha > 2/3$ (averaging the dilations along u), and the present author's extension of that theorem, recorded in the revised part III, gives $s > 0$ for $\alpha > 1/2$ by a dispersion argument with Weil's bound. Below $1/2$ both mechanisms fail: the

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incomplete Kloosterman sums that dispersion produces have modulus $[d_1, d_2] \asymp Y^2$, and Weil's bound is trivial once the range of u is shorter than the square root of that modulus. This paper supplies two further mechanisms, one on each side of $\alpha = 1/3$.

Results. Theorem 5 treats α near $1/2$: the weight w is decomposed as $1 * \chi_D * \gamma$, and the resulting bilinear forms in u are estimated by Weil's bound when one factor is long (Type I) or short (Type II), and, in the remaining balanced case, by the bilinear bound of Bourgain and Garaev [1] for Kloosterman fractions. That bound needs an estimate for the additive energy of reciprocals; the one in [1] is too weak at the sizes that occur here, and Lemma 2 supplies a stronger one, valid for every modulus, by transferring an argument of Cilleruelo and Garaev [2] for points on modular hyperbolas. Theorem 7 treats small α for negative discriminants: piece u is, after removing the content, a Type I sum for the polynomial $u^2 \ell^2 + |D|$ in the sense of Grimmelt and Merikoski [3], and their automorphic kernel theorem [4] applies at level $u^2 e$ once the two kernel quantities are estimated with the dilation tracked. The diagonal is the number of orbits of the family, $\asymp u$; the off-diagonal, which their general bound would give as $\asymp u^2$, is in fact bounded independently of u (Lemma 6), because the distances inside the family are quantised in steps of $1/(4h)$ whatever the dilation.

What remains. Both mechanisms degenerate at $\alpha = 1/3$, from opposite sides and for unrelated reasons: there the balanced Type II atoms sit at $d^{1/4}$, where the energy input is not strong enough, and the kernel method's diagonal $\asymp u$ meets the trivial bound. The final section explains what the residual band costs. Two points are worth recording. A band of width 2δ in α costs $\delta \cdot H(\log H)^{1-c} \log \log H$, not $\delta H \log H$, because part III's Theorem E applies piece by piece inside it; so a failing method never makes the total worse. And if the savings on both sides vanish only at $\alpha = 1/3$, with implied constants that are powers of $\log H$, then the blocks near the point contribute $O(H)$ in total, which would give $\text{Off}_f^*(H) = O(H)$ — Hypothesis (E) of part I [5], a bounded second-order term in the Cesàro asymptotic. The constant A_f needs strictly more: a saving at the point itself.

2 Notation

We follow part III [6]. For a prime p let $\omega(p)$ be the number of roots of f modulo p , $\lambda(p) = p/(p - 2\omega(p))$, $P_p = 1 - \omega(p)^2/(p - \omega(p))^2$, $P(1) = \prod_p P_p$ and $w(u) = P(1)\lambda(u)\omega(u)$; w is supported on squarefree u and $\sum_{u \leq t} |w(u)| \ll t$. A modulus d' is *admissible* if it is squarefree, all its prime factors are split, and $(d', 2Du) = 1$; $R_{d'}^{(u)} = \{x \bmod d' : u^2 x^2 \equiv D\}$, $\rho_k(d'; a) = \sum_{r \in R_{d'}} e(kar/d')$, and $\langle x \rangle_{d'} \in [1, d']$ is the representative of x . With $B_{d'}^{(Y)}(m) = \sum_{h \leq Y, h \equiv m \pmod{d'}} (Y - h) - Y^2/(2d')$, the piece is

$$\mathcal{P}_u(Y) = \sum_{d' > 1 \text{ admissible}} \frac{\lambda(d')}{d'} \sum_{x \in R_{d'}^{(u)}} B_{d'}^{(Y)}(\langle x \rangle_{d'}), \quad Y = H/u.$$

We write $G_{d'}(a, Y) = \sum_{r \in R_{d'}} B_{d'}^{(Y)}(\langle ar \rangle_{d'})$, $\bar{G}_{d'}(Y)$ for its mean over the units a , $G^\circ = G - \bar{G}$, and $\xi_k(d', N) = (1 - \cos(2\pi kN/d'))/\sin^2(\pi k/d')$ for the coefficients of the finite Fourier expansion of $B^{(N)}$ (part III, Lemma finfourier). Throughout, x^* denotes the inverse of x modulo the modulus at hand, $e_m(z) = e(z/m)$, and $\varepsilon > 0$ is arbitrarily small. Parameters: $\delta = 10^{-3}$, $\eta = \delta/10$, $\delta_3 = \delta$, $\varepsilon' = \eta/7$, $\kappa = \delta/2$, $K_1 = H^{2\eta}$.

3 The energy of reciprocals and bilinear Kloosterman fractions

We write x^* for the inverse of x modulo the modulus at hand and $e_m(z) = e(z/m)$.

Lemma 1 (smooth incomplete Kloosterman sums). *Let $q \geq 1$ be squarefree, $a \in \mathbb{Z}$, $X > 0$ and $T \geq 1$, and let W be smooth, supported in an interval of length at most $4X$, with $\|W\|_\infty \ll 1$ and $\int |W''| \ll T^2/X$. Then*

$$\sum_{(v,q)=1} W(v) e_q(av^*) \ll \tau(q) \left(1 + \frac{X}{q}\right) T^2 q^{1/2} (a, q)^{1/2}.$$

Proof. Split v into classes modulo q and apply Poisson summation in each: the sum equals $\sum_h \widehat{W}(h/q) S(a, h; q)/q$ with $S(a, h; q)$ the Kloosterman sum. For squarefree q , Weil's bound at the primes and twisted multiplicativity give $|S(a, h; q)| \leq \tau(q) (a, h, q)^{1/2} q^{1/2}$, which covers $h = 0$, where $S(a, 0; q) = c_q(a)$ is a Ramanujan sum and $|c_q(a)| \leq (a, q) \leq q^{1/2} (a, q)^{1/2}$. Since $\widehat{W}(\xi) \ll \min(X, T^2/(X\xi^2))$, one has $q^{-1} \sum_h |\widehat{W}(h/q)| \ll T^2(1 + X/q)$. $\square$

Lemma 2 (additive energy of reciprocals, every squarefree modulus). *Let $m \geq 2$ be squarefree, $1 \leq N \leq m$, and $J_4(N) = \#\{x \in [1, N]^4 : (x_i, m) = 1, x_1^* + x_2^* \equiv x_3^* + x_4^* \pmod{m}\}$. Then for every $\varepsilon > 0$,*

$$J_4(N) \ll_\varepsilon m^\varepsilon (N^2 + N^{7/2} m^{-1/2}).$$

Proof. *Step 1: a hyperbola in a square box.* Let $(\lambda, m) = 1$, $K \in \mathbb{Z}$ and $1 \leq M \leq m$; we claim $I := \#\{(x, y) \in (K, K+M]^2 : xy \equiv \lambda \pmod{m}\} \ll_\varepsilon m^\varepsilon (M^{3/2} m^{-1/2} + 1)$. For $M = 1$ this is trivial, so let $M \geq 2$. Writing $x = K + x'$, $y = K + y'$ with $1 \leq x', y' \leq M$, the congruence becomes $x'y' + Kx' + Ky' \equiv b := \lambda - K^2 \pmod{m}$. Put $T = \lfloor (m/M)^{1/2} \rfloor \geq 1$; by Dirichlet's theorem there are $1 \leq t \leq T$ and u_0 with $tK \equiv u_0 \pmod{m}$ and $|u_0| \leq \min(m/T, m/2)$. Multiplying by t ,

$$tx'y' + u_0x' + u_0y' = b_0 + zm, \quad |b_0| < m/2, \quad |z| \leq \frac{TM^2}{m} + \frac{2M}{T} + \frac{1}{2} \leq 5M^{3/2} m^{-1/2} + \frac{1}{2}.$$

For each z this is $(tx' + u_0)(ty' + u_0) = n_z := t(b_0 + zm) + u_0^2$, and $|n_z| \leq 7m^2$. Now $n_z \equiv t^2(x' + K)(y' + K) \equiv t^2\lambda \pmod{m}$, and λ is a unit with $1 \leq t^2 \leq m/M < m$, so $m \nmid n_z$ and in particular $n_z \neq 0$. Each z therefore contributes at most $\tau(|n_z|) \ll_\varepsilon m^\varepsilon$ pairs, and summing over the $\ll M^{3/2} m^{-1/2} + 1$ admissible z gives the claim.

Step 2: from energy to hyperbolas. Let $J_2(\lambda) = \#\{x, y \leq N, (xy, m) = 1 : x^* + y^* \equiv \lambda\}$, so $J_4 = \sum_\lambda J_2(\lambda)^2$ and $\sum_\lambda J_2(\lambda) = N'^2 \leq N^2$ with $N' = \#\{x \leq N : (x, m) = 1\}$. For a unit λ the congruence $x^* + y^* \equiv \lambda$ is equivalent to $(x - \lambda^*)(y - \lambda^*) \equiv \lambda^{*2}$, a unit hyperbola in a square box with equal shifts, so Step 1 gives $J_2(\lambda) \ll m^\varepsilon (N^{3/2} m^{-1/2} + 1)$. If $g = \gcd(\lambda, m) > 1$ then $p \mid g$ forces $x \equiv -y \pmod{p}$, hence $x \equiv -y \pmod{g}$ and $g \leq 2N$; modulo m/g the residue λ is a unit, so Step 1 applied there gives $J_2(\lambda) \ll m^\varepsilon (g^{1/2} N^{3/2} m^{-1/2} + 1)$ when $N \leq m/g$, while $J_2(\lambda) \leq N$ always. Using $\sum_{\gcd(\lambda, m)=g} J_2(\lambda) \leq N(2N/g + 1)$ and summing over the divisors $g \leq 2N$ of m yields the lemma; in the range $N > m/g$ one uses $4N^4/m \leq 4N^{7/2} m^{-1/2}$. $\square$

Lemma 3 (Bourgain–Garaev). *Let $m \geq 2$, $(a, m) = 1$, $1 \leq N_1, N_2 \leq m$, and let α_1, α_2 be coefficients bounded by 1 and supported on units. Then for all $k_1, k_2 \geq 1$,*

$$\left| \sum_{x_1 \leq N_1} \sum_{x_2 \leq N_2} \alpha_1(x_1) \alpha_2(x_2) e_m(ax_1^* x_2^*) \right|^{2k_1 k_2} \leq m N_1^{2k_1 k_2 - 2k_1} N_2^{2k_1 k_2 - 2k_2} J_{2k_1}(N_1) J_{2k_2}(N_2).$$

Proof. This is the inequality established in the proof of Theorem 3 of [1]: two applications of Hölder's inequality followed by Cauchy–Schwarz, using $\sum_\lambda J_k(\lambda) \leq N^k$ and that a is a unit. $\square$

Corollary 4. *For squarefree m and $m^{1/3} \leq N_1, N_2 \leq m$, under the hypotheses of Lemma 3, $|\sum \sum \alpha_1 \alpha_2 e_m(ax_1^* x_2^*)| \ll_\varepsilon m^\varepsilon (N_1 N_2)^{15/16}$.*

Proof. Take $k_1 = k_2 = 2$ in Lemma 3 and insert Lemma 2: for $N_i \geq m^{1/3}$ the second term of the energy bound dominates, $N_i^{7/2} m^{-1/2} \geq N_i^2$, so the right-hand side is $\ll m^{1+4\varepsilon} (N_1 N_2)^4 (N_1 N_2)^{7/2} m^{-1} = m^{4\varepsilon} (N_1 N_2)^{15/2}$. $\square$

4 The pieces around $u = H^{1/2}$

Theorem 5. *Let f be as above. There is $\delta_M > 0$ (any $\delta_M < 4\eta/7$ with the parameters above, so $\delta_M = 5 \cdot 10^{-5}$ is admissible) such that*

$$\sum_{H^{0.47} < u \leq H^{0.53}} w(u) \mathcal{P}_u(H/u) \ll_f H^{1-\delta_M}.$$

Consequently, with part III's Theorem on the range $u > H^{1/2+\varepsilon}$ applied at $\varepsilon = 0.03$, $\sum_{u > H^{0.47}} w(u) \mathcal{P}_u(H/u) = c_{\text{off}}(f)H + O(H^{1-\delta_M})$.

[Full proof in research/explore/PAPER-V-thmM.md (with REVISION 1 and REVISION 2 merged); three independent readings, recorded in PROOFS-uniform.md §28, §30, §35, the last confirming the merged text. What follows is the skeleton to be expanded into the paper text.]

Structure of the proof. Fix a smooth dyadic partition of the range of u , with ramps of relative width $H^{-\kappa}$ at the two ends; the transition zones are $\ll H^{1-\kappa}(\log H)^C$ by the trivial bound $\mathcal{P}_u(Y) \ll Y$. For one block $U \in [H^{0.47}/4, 2H^{0.53}]$, $Y = H/U$, $a = \log U / \log H \in [0.46, 0.54]$:

- R1. *Reduction.* The dilation average contributes $\ll U(\log H)^C$; the moduli $d' > YH^\eta$ are part III's Step 3, $\ll H^{1-\eta+3\varepsilon'}$; the moduli $d' \leq Y^{1-\delta}$ are $\ll UY^{1-\delta}$ by Shiu's theorem; on the remaining range the finite Fourier expansion applies, the frequencies $k > K_1$ cost $H^{1-\eta}$, and a Mellin transform separates the coefficients $\xi_k(d', H/u)$ from u , at a cost of $\ll Y(\log H)^3$ after summing over k and d' .
- R5. *Decomposition.* $w = c_f(1 * \chi_D * \gamma)$ with $\sum_n |\gamma(n)| n^{-1/2-\varepsilon} < \infty$; the factor $n_3 > H^{\delta_3}$ is removed trivially; the atoms n_1 (weight 1) and n_2 (weight χ_D) are split into smooth dyadic boxes. Three cases exhaust the boxes.
- R2. *Type I:* an atom of length $N \geq Y^{1/2} H^{3\delta}$ gives, by Lemma K applied for each modulus and cofactor, a saving $H^{-1.79\delta}$.
- R3. *Type II:* an atom of length $U_2 \in [H^{3\delta}, UY^{-1/2} H^{-3\delta}]$ gives, by Cauchy–Schwarz over (d', u_1) and Weil's bound for the resulting incomplete Kloosterman sums, a saving $H^{-1.45\delta}$.
- R4. *The residual:* both atoms of size between $UY^{-1/2} H^{-3\delta}$ and $Y^{1/2} H^{3\delta}$. Then both exceed $m^{1/3}$ for $m = d'/(k, d')$, and Corollary 4 gives a saving $H^{-0.028}$.

Summing the three savings against the $O(\log H)$ blocks, the binding constraint is part III's Step 3, which yields $\delta_M < 4\eta/7$.

5 The small dilations

Lemma 6 (the off-diagonal at one level does not grow with the dilation). *Let $h \geq 1$, u squarefree with $(u, 2h) = 1$, $a = u^2$, and let F_u be the set of positive definite forms $g = \begin{pmatrix} A & B \\ B & C \end{pmatrix}$ with $AC - B^2 = ah$, $a \mid B$, $a \mid C$. For $Z \geq 1$,*

$$\sum_{g_2 \in \Gamma_0(a) \backslash F_u} \sum_{\substack{g_1 \in F_u, \, g_1 \neq g_2 \\ u(g_1, g_2) \leq Z}} (1 + u(g_1, g_2))^{-1/2} \ll_{\varepsilon} (hZu)^{\varepsilon} h^{O(1)} (1 + Z^{1/2}),$$

uniformly in u , where $u(\cdot, \cdot)$ is the point-pair invariant.

[Proof in PAPER-V-smallend-draft.md (undilation to $\Gamma_0^0(u)$, lift to level one, a local count of common orthogonal lines, and a sum over the quantised distances); under reading.]

Theorem 7. *Let f be as above with $D < 0$. For every $c > 0$ there is $\delta_S(c) > 0$ with $\sum_{u \leq H^{1/3-c}} w(u) \mathcal{P}_u(H/u) \ll_f H^{1-\delta_S(c)}$.*

[Proof outline S1–S8 in PAPER-V-smallend-draft.md, resting on [4] Theorem 8.1 and [3] Lemma 3.1, with Lemma 6 and the diagonal count; $\delta_S(c) \asymp c$ once the exponent in the kernel theorem's $\delta^{-O(1)}$ is named. Under reading.]

6 What remains: the band at $u = H^{1/3}$

Theorems 5 and 7 leave a neighbourhood of $\alpha = 1/3$ uncovered, and the two mechanisms fail there for unrelated reasons. On the upper side the balanced Type II atoms sit at $d^{1/4}$, where the available energy input for reciprocals is exactly at its threshold: what would be needed is a bound for the six-fold energy $J_6(N)$ below N^4 at $N \asymp m^{1/4}$, which is not known even for prime moduli. On the lower side the kernel method's diagonal, the number of orbits of the family, is $\asymp u$, and at $u \asymp Y^{1/2}$ it meets the trivial bound.

Three remarks fix what such a band costs, and what closing it would buy.

Remark 8 (a band is cheap, and a failing method is free). Inside any range of α one still has part III's Theorem E piece by piece, $\mathcal{P}_u(H/u) \ll (H/u)(\log H)^{-c} \log \log H$ uniformly in $u \leq H^{1-\varepsilon}$. Hence a band $[H^{\alpha_0-\delta}, H^{\alpha_0+\delta}]$ on which nothing better is available costs $\ll_f \delta \cdot H(\log H)^{1-c} \log \log H$, not $\delta H \log H$: a mechanism that degenerates can never make the total worse than what is already proved, and any band whose width tends to 0 improves Theorem E by that factor.

Remark 9 (a point gap costs $O(H)$). Suppose the savings on the two sides vanish only at $\alpha = 1/3$, say like $H^{-c|\alpha-1/3|}$, with implied constants that are powers of $\log H$. Summing over dyadic blocks, $\sum_k H e^{-c|k \log 2 - \frac{1}{3} \log H|} = H(1 + 2/(e^{c \log 2} - 1)) = O(H/c)$: the blocks near the point contribute $O(H)$ in total. Since the main term $c_{\text{off}}(f)H$ comes from $u > H^{1/2+\varepsilon}$ and the remaining blocks are error terms, this would give $\text{Off}_f^*(H) = O(H)$, which is Hypothesis (E) of part I [5] and yields $\sum_{h \leq H} (1 - h/H)(S_f(h) - C(f)^2) = -\frac{1}{2}C(f) \log H + O_f(1)$: a bounded second-order term.

Remark 10 (what A_f needs). The constant of Conjecture 1 of part I requires $\text{Off}_f^*(H) = c_{\text{off}}(f)H + o(H)$, hence a saving factor $\gg \log H$ uniformly on $|\alpha - 1/3| \ll \log \log H / \log H$; a fixed positive power saving at $\alpha = 1/3$ itself suffices. A linearly vanishing saving, however small the rate, gives $O(H)$ but never $o(H)$.

Remark 11 (what is actually binding). An audit of every implied constant in the chain (research/explore/PROOFS-uniform.md §36) shows that no divisor-size loss constrains the

argument: the divisor factors all occur under sums and average to powers of $\log H$, the kernel theorem's $\delta^{-O(1)}$ is δ^{-20} , its $(AD)^{o(1)}$ is $(\log AD)^{O(1)}$, and its $q^{o(1)}$ is 1 in the present application. What binds is the constant in Bourgain and Garaev's Hölder step, which grows as $k_1 \asymp 2/(9|\alpha - 1/3|) \rightarrow \infty$: the band on which the estimate is effective has width $\asymp (\log \log H / \log H)^{1/3}$. With the in-band fallback on part III's Theorem E this gives, once the map is written out, $\text{Off}_f^*(H) \ll H(\log H)^{0.63+o(1)}$ for $D < 0$, while everything outside the band sums to $O(H)$. The step from there to $\text{Off}_f^*(H) = O(H)$ therefore needs one thing only: an estimate that survives at the exceptional sizes $N \asymp m^{1/4}$ — an energy bound J_6 below N^4 for composite moduli, or a different tool there. A level-averaged spectral bound for the twist $e(c\bar{r}d/x)$ with $x \asymp Y^{1/2}$ would serve the same purpose; neither is in the literature.

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